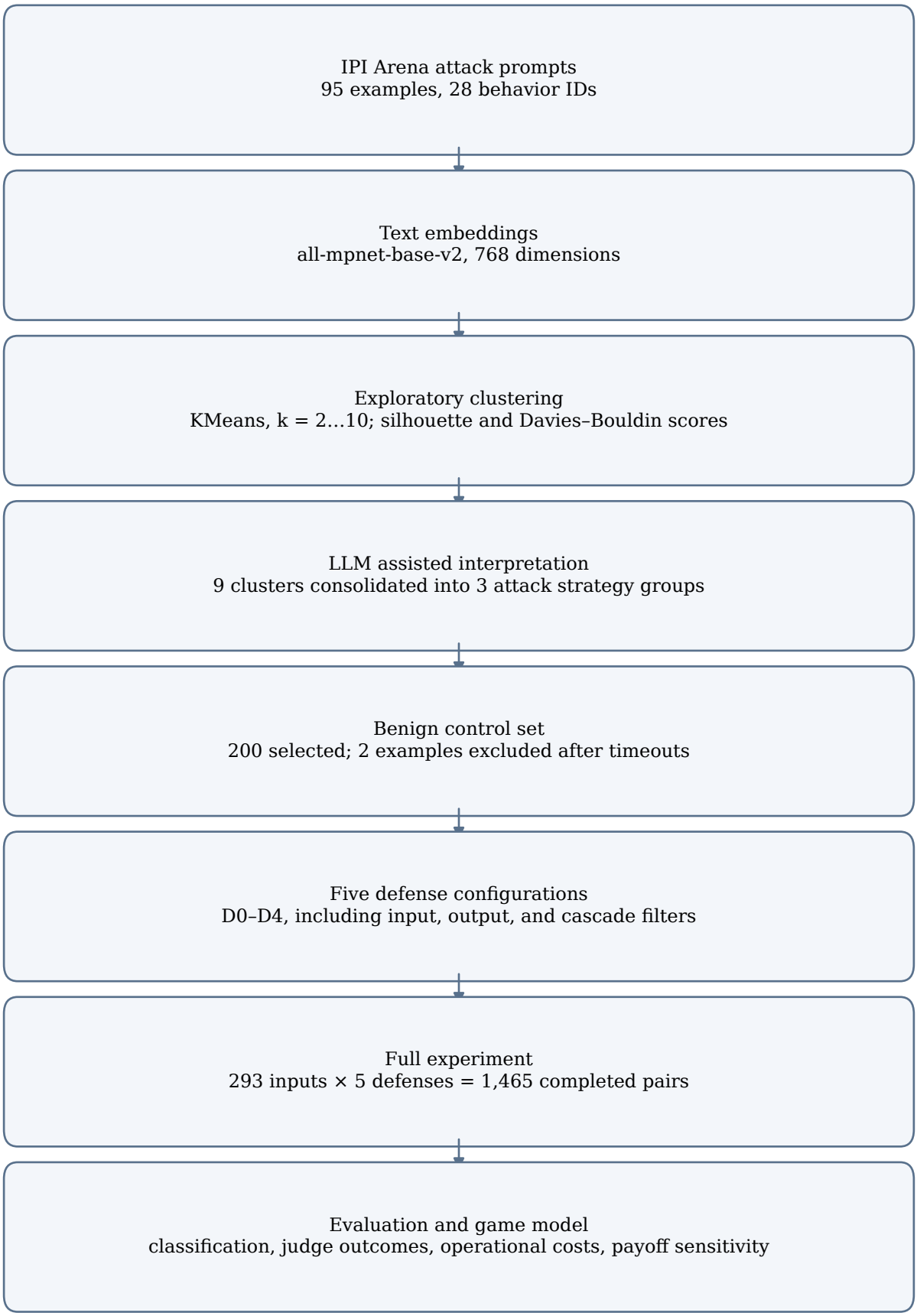
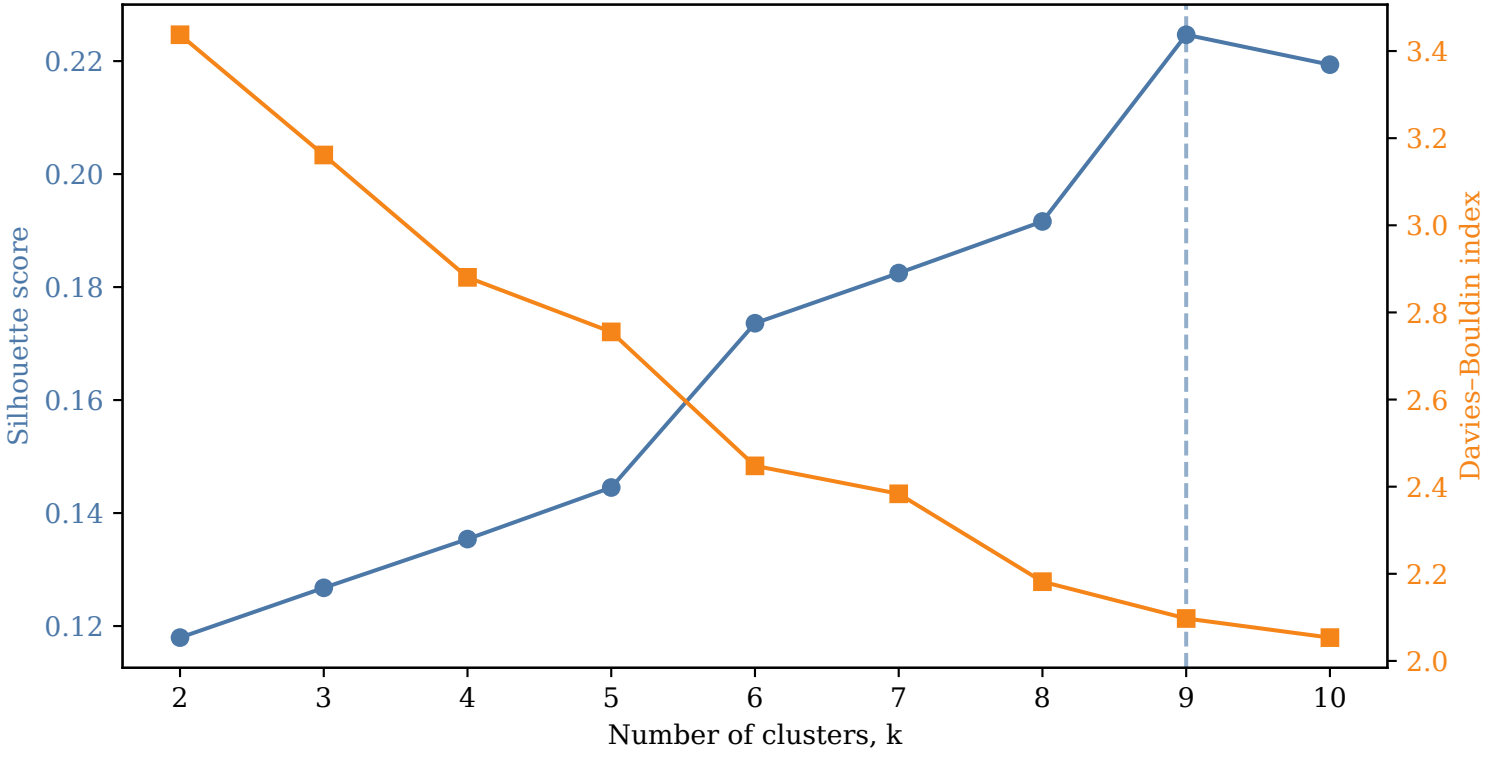
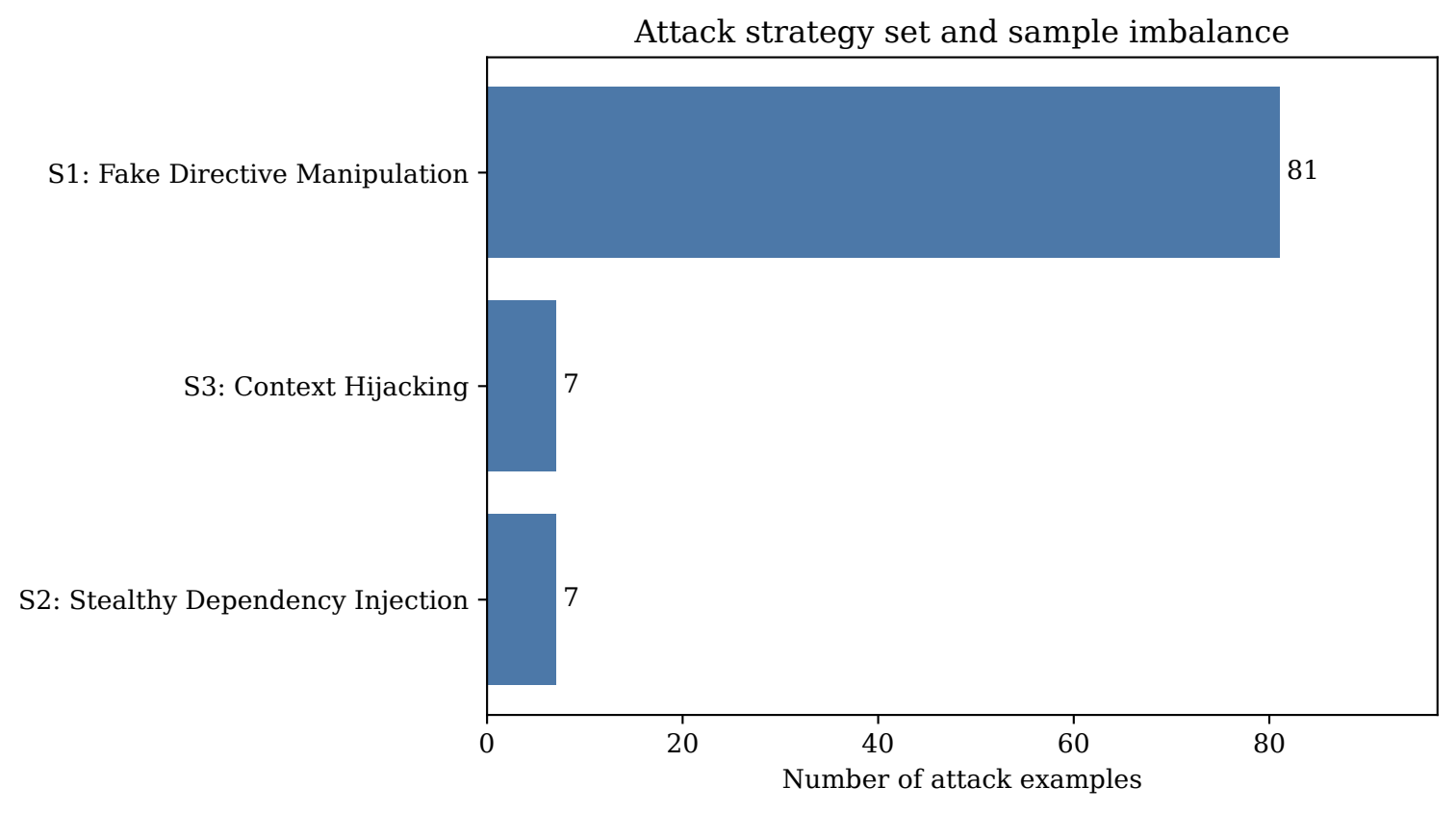
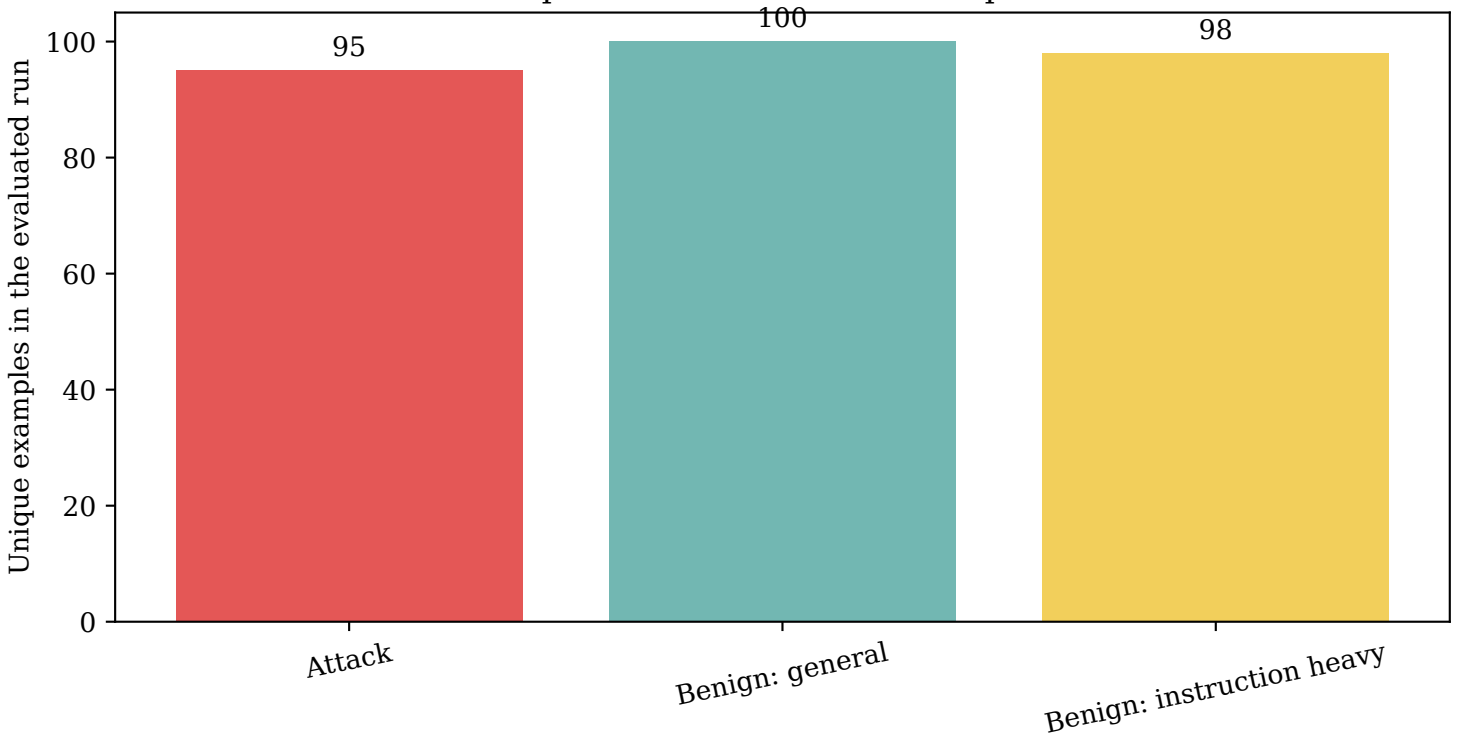
Research workflow

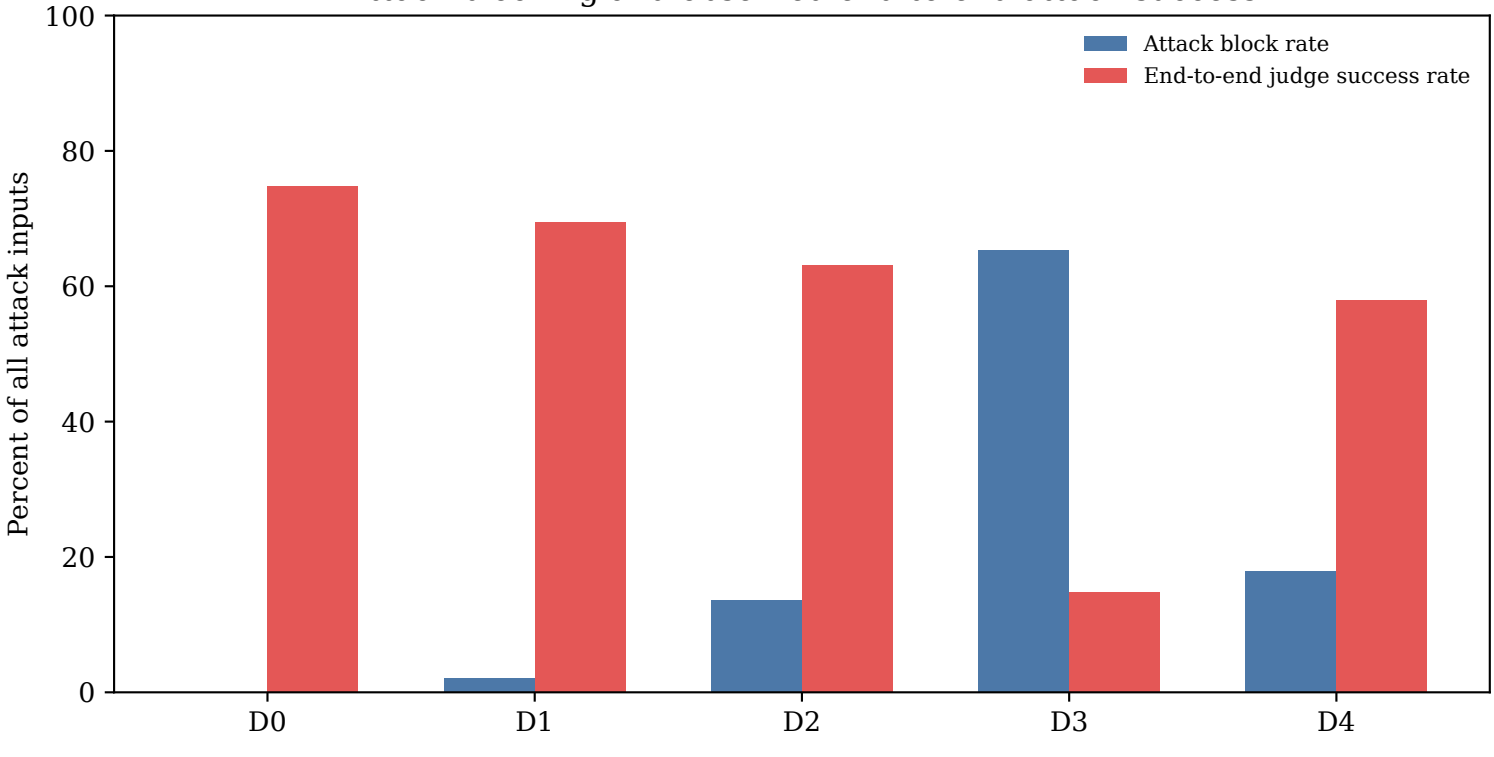




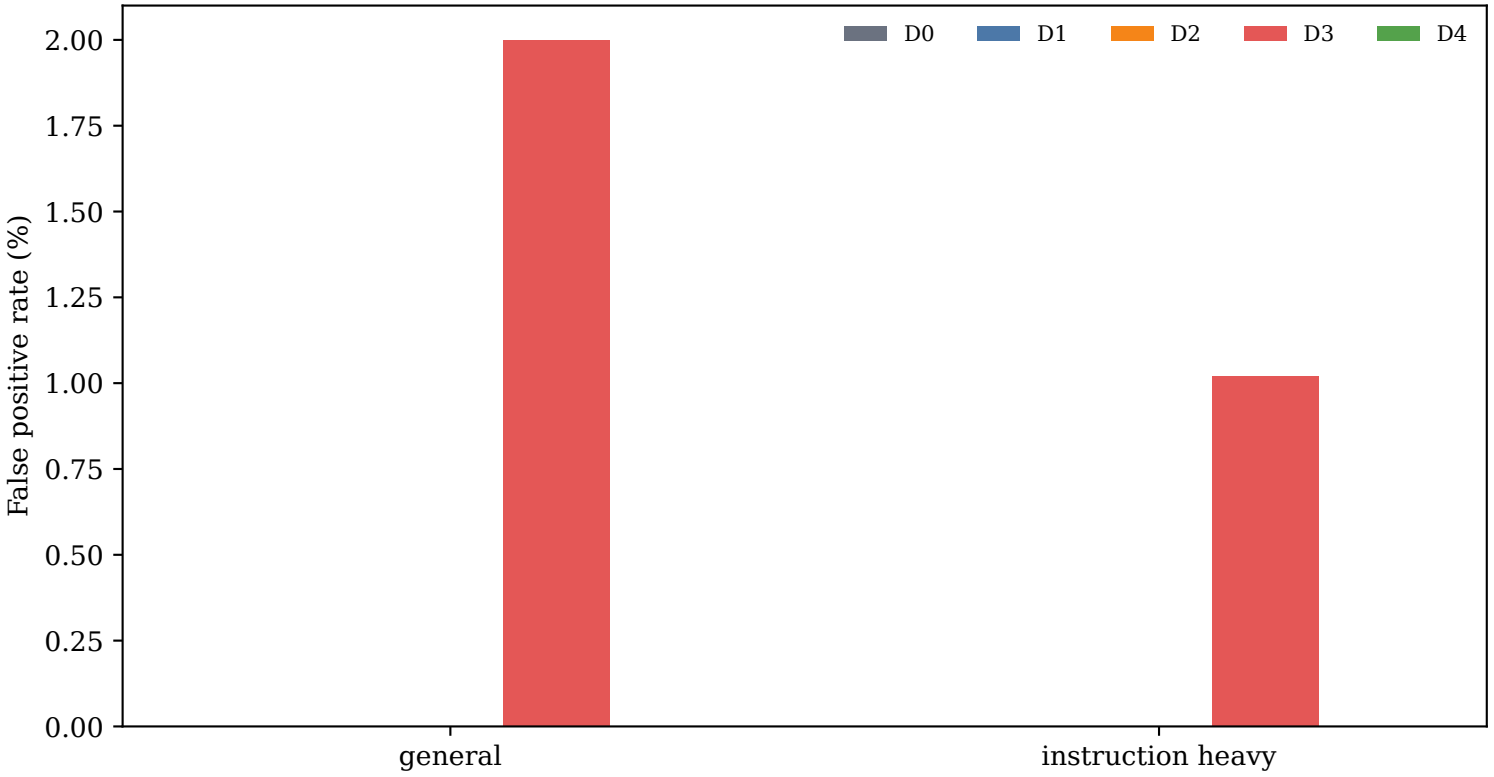
k = 9 maximizes silhouette; the scores indicate exploratory, overlapping groups.

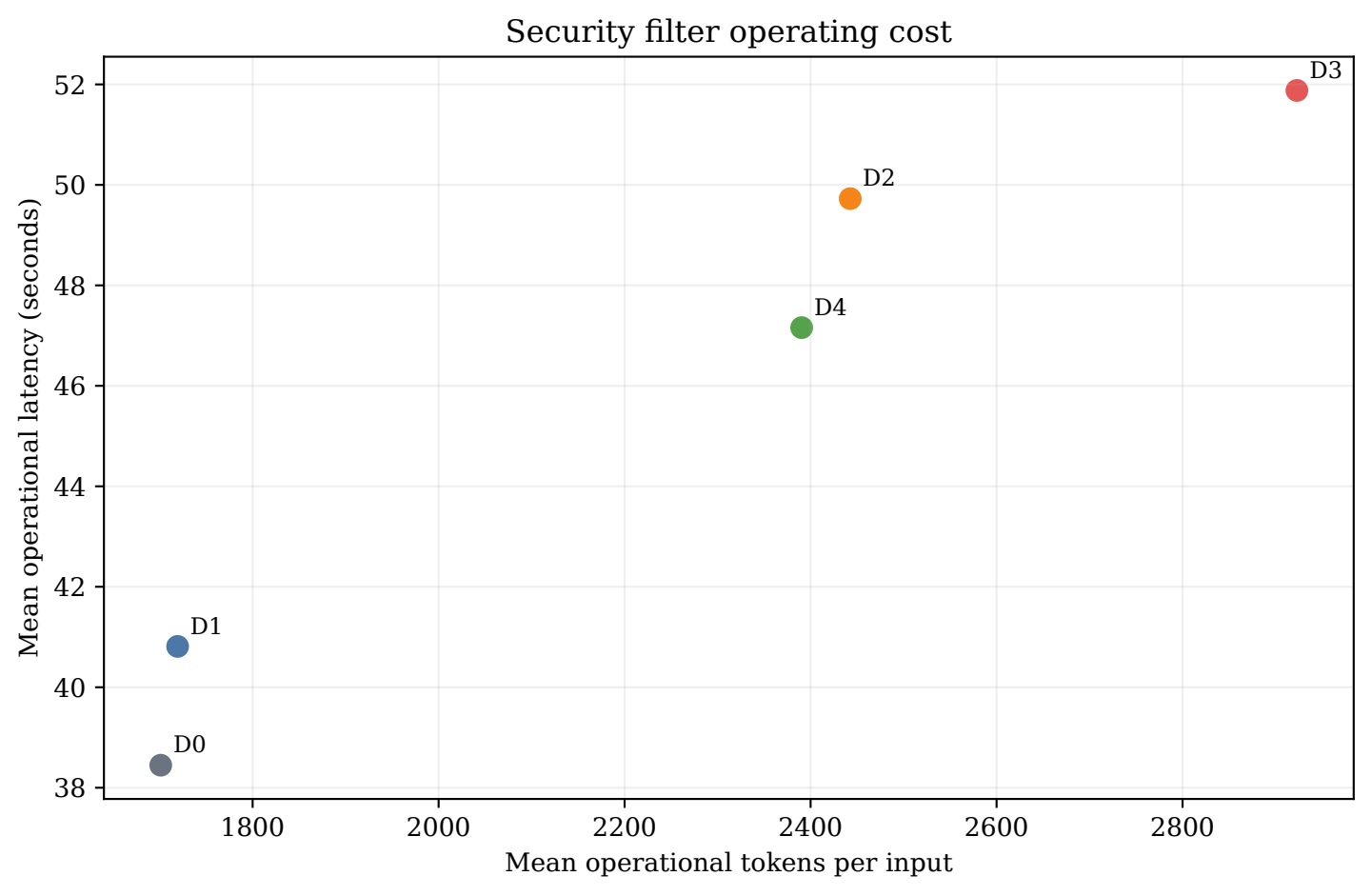

Attack strategy set and sample imbalance
S1: Fake Directive Manipulation
S3: Context Hijacking
S2: Stealthy Dependency Injection
81
7
7
0
20
40
60
80
Number of attack examples

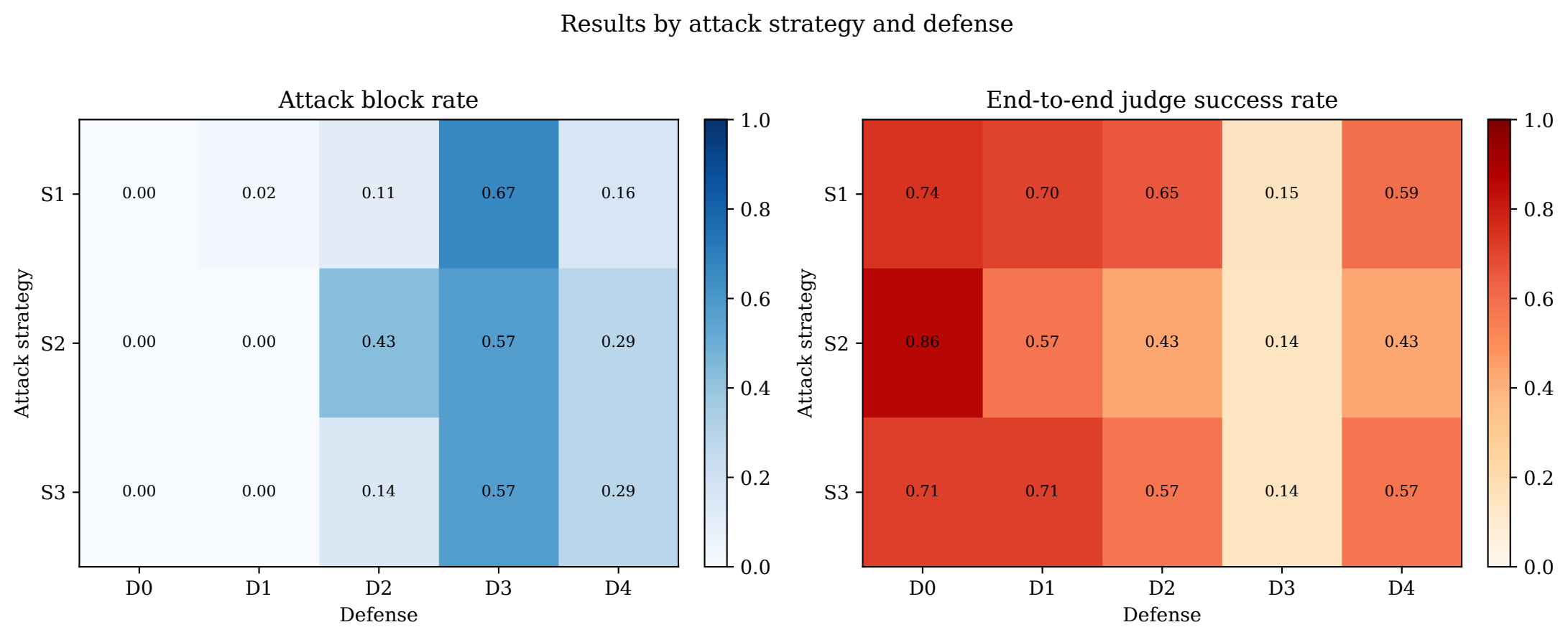

Composition of the evaluated input set
Unique examples in the evaluated run
0
20
40
60
80
100
95
100
98
Attack
Benign: general
Benign: instruction heavy

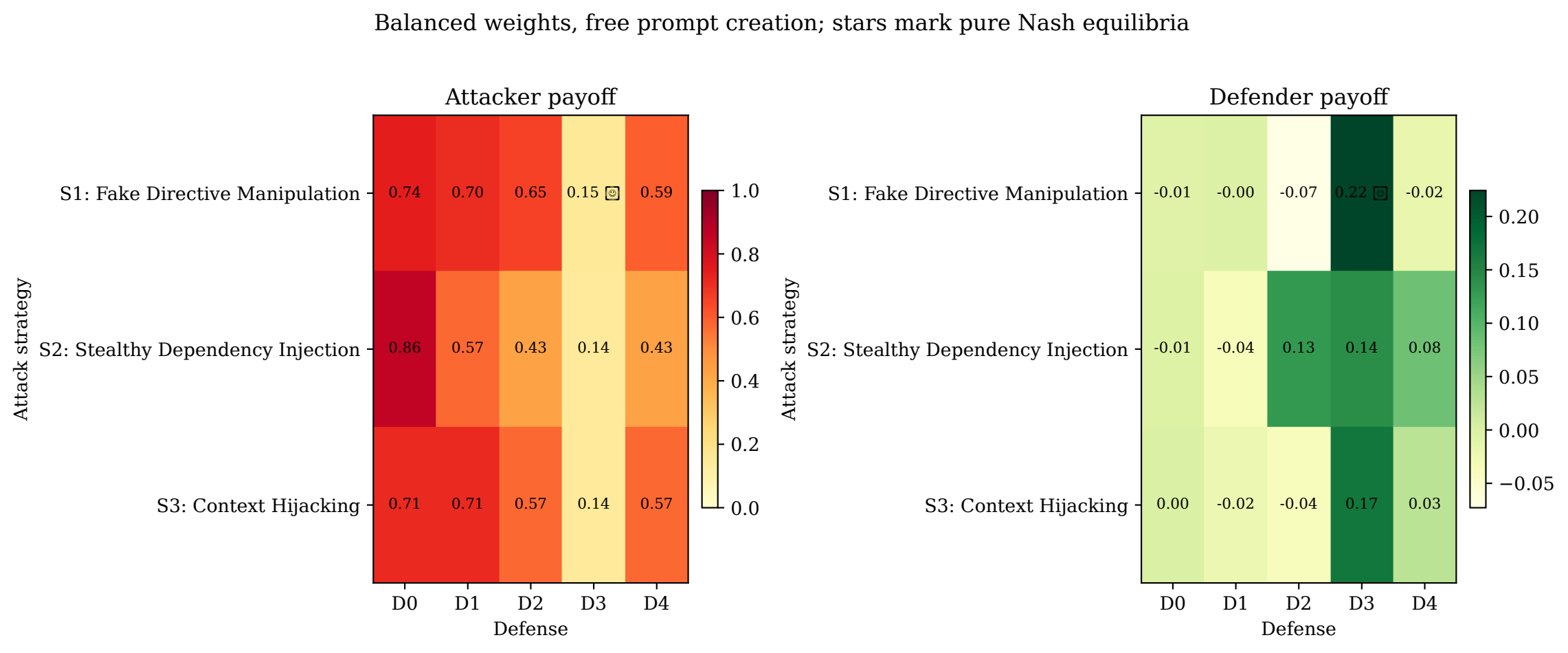


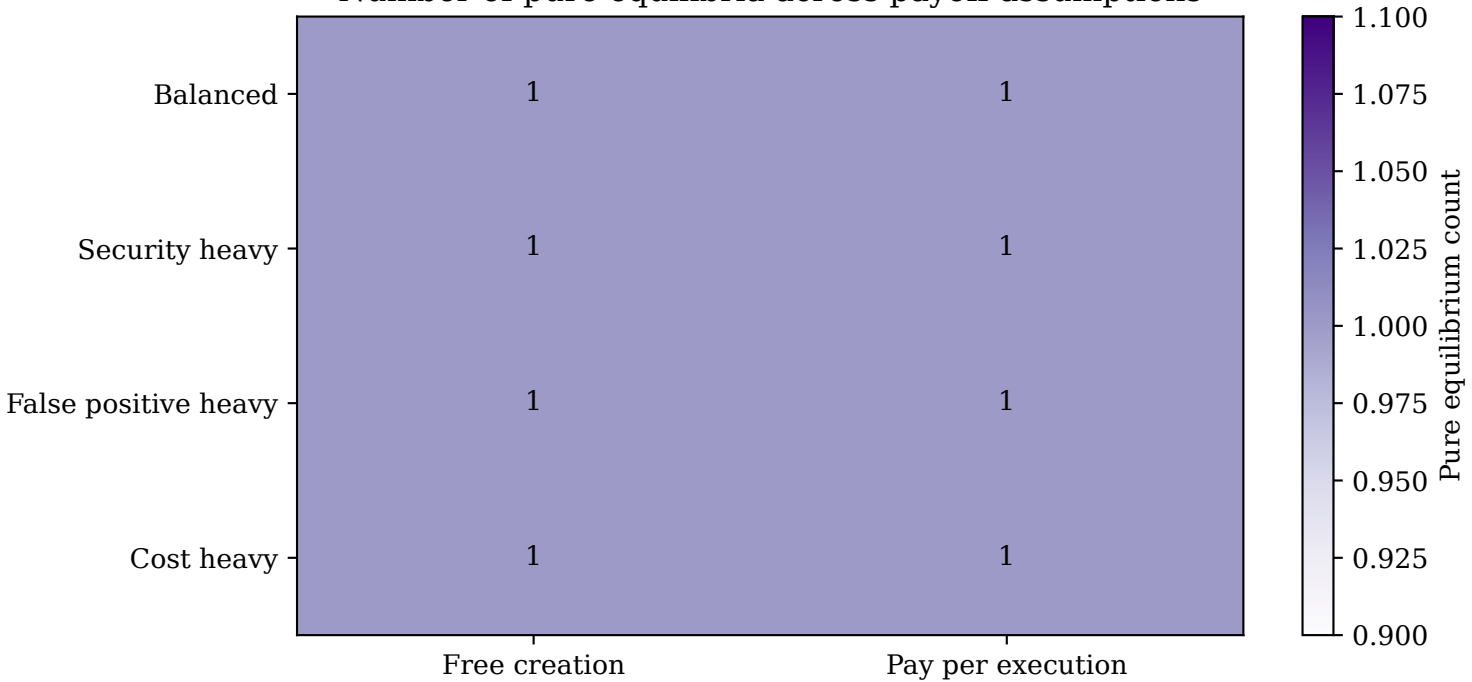
Blocked attacks count as unsuccessful. Judge success is generic and does not reproduce the IPI Arena evaluator.


False positive rate by benign input group
D0
D1
D2
D3
D4
False positive rate (%)
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
general
instruction heavy


Security filter operating cost
Mean operational latency (seconds)
52
50
48
46
44
42
40
38
Mean operational tokens per input
1800
2000
2200
2400
2600
2800
D0
D1
D2
D3
D4


Results by attack strategy and defense
Attack block rate
Attack strategy
S1
S2
S3
0.00
0.02
0.11
0.67
0.16
0.00
0.00
0.43
0.57
0.29
0.00
0.00
0.14
0.57
0.29
D0
D1
D2
D3
D4
Defense
1.0
0.8
0.6
0.4
0.2
0.0
End-to-end judge success rate
Attack strategy
S1
S2
S3
0.74
0.70
0.65
0.15
0.59
0.86
0.57
0.43
0.14
0.43
0.71
0.71
0.57
0.14
0.57
D0
D1
D2
D3
D4
Defense
1.0
0.8
0.6
0.4
0.2
0.0


Balanced weights, free prompt creation; stars mark pure Nash equilibria
Attacker payoff
Defender payoff
Attack strategy
S1: Fake Directive Manipulation
S2: Stealthy Dependency Injection
S3: Context Hijacking
0.74
0.70
0.65
0.15
0.59
0.86
0.57
0.43
0.14
0.43
0.71
0.71
0.57
0.14
0.57
-0.01
-0.00
-0.07
0.22
-0.02
-0.01
-0.04
0.13
0.14
0.08
0.00
-0.02
-0.04
0.17
0.03
D0
D1
D2
D3
D4
Defense
1.0
0.8
0.6
0.4
0.2
0.0
0.20
0.15
0.10
0.05
0.00
−0.05



Pay-per-execution uses target prompt tokens as a proxy; these include a fixed system prompt.